1 When do four points lie on a circle?

Points that lie on one circle are called **concyclic**. Any three points that are not on one line are concyclic (Theorem 5). What about four points?

Solve first.

P1. Draw a rough figure for each part.

(a) $ABCD$ is a rectangle with diagonals meeting at O. Explain why $OA = OB = OC = OD$. Are A, B, C and D concyclic?

(b) Draw a rhombus $PQRS$ with side 4 cm and $\angle P = 60°$. Draw the circle through P, Q and S. Does it pass through R?

2 Equal angles on the same side

Solve first.

P2. A, B, C are points on a circle. D is a point on the same side of AB as C, and $\angle ADB = \angle ACB$. Suppose D is *outside* the circle. Then AD cuts the circle at a point E between A and D.

(a) Why is $\angle AEB = \angle ACB$?

(b) $\angle AEB$ is an exterior angle of $\triangle BED$. Which is larger, $\angle AEB$ or $\angle ADB$?

(c) Use (a), (b) and $\angle ADB = \angle ACB$. What goes wrong?

(d) What goes wrong if D is *inside* the circle? (Extend AD to meet the circle at E.)

(e) So where must D be?

> **Theorem 10** (NCERT Theorem 10). *If a line segment AB joining two points A, B subtends equal angles at two other points C, D that lie on the same side of AB, then the four points lie on a circle.*

FIG

Why it works.

Given: C and D lie on the same side of AB, and $\angle ACB = \angle ADB$.
To show: A, B, C and D lie on one circle.

1. A, B and C are not on one line, so one circle passes through them (Theorem 5).
2. If D were outside this circle, P2(c) shows that $\angle ACB > \angle ACB$. If D were inside, P2(d) gives the same kind of contradiction.
3. So D lies on the circle.

Theorem 10 is the converse of "angles in the same segment are equal".

Example 1. C and D lie on the same side of AB, with $\angle ACB = \angle ADB = 40°$ and $\angle CAD = 25°$. Find $\angle CBD$.

FIG

Given: $\angle ACB = \angle ADB = 40°$, C and D on the same side of AB, $\angle CAD = 25°$.
To find: $\angle CBD$.

1. A, B, C and D are concyclic (Theorem 10).
2. $\angle CBD$ and $\angle CAD$ stand on the chord CD, in the same segment. So $\angle CBD = \angle CAD = 25°$.

Answer: $\angle CBD = 25°$.

3 Cyclic quadrilaterals

A quadrilateral whose four vertices lie on a circle is a **cyclic quadrilateral**.

Solve first.

P3. $ABCD$ is a cyclic quadrilateral in a circle with centre O. The arc BCD subtends $\angle BOD = 140°$ at O.

(a) A is outside the arc BCD. Find $\angle BAD$.
(b) What angle does the arc BAD subtend at O? Find $\angle BCD$.
(c) Find $\angle BAD + \angle BCD$.

P4. Repeat P3 with $\angle BOD = 100°$. What do you notice?

P5. In P3, find $\angle ABC + \angle ADC$ without any more information.

Theorem 11 (NCERT Theorem 11). *The sum of two opposite angles of a cyclic quadrilateral is* 180°.

FIG

Why it works.

Given: $ABCD$ is a cyclic quadrilateral in a circle with centre O.
To show: $\angle BAD + \angle BCD = 180°$.

1. The arc BCD subtends an angle p at O, and the arc BAD subtends the rest of the complete angle, $q = 360° - p$.
2. A is outside the arc BCD, so $\angle BAD = \frac{p}{2}$. C is outside the arc BAD, so $\angle BCD = \frac{q}{2}$ (Theorem 9).
3. $\angle BAD + \angle BCD = \frac{p+q}{2} = \frac{360°}{2} = 180°$.

The other pair also adds up to 180°, because the four angles add up to 360° (P5).

Solve first.

P6. $ABCD$ is a quadrilateral with $\angle BAD + \angle BCD = 180°$. Draw the circle through A, B and D. Suppose C is *outside* this circle, and CD cuts the circle at E, between D and C.

(a) $ABED$ is a cyclic quadrilateral. Find $\angle BAD + \angle BED$.

(b) Compare $\angle BED$ with $\angle BCD$.

(c) $\angle BED$ is an exterior angle of $\triangle BEC$. What goes wrong?

(d) So where must C be?

Theorem 12 (NCERT Theorem 12). *If two opposite angles of a quadrilateral add up to* 180°*, then the vertices of the quadrilateral lie on a circle, that is, they are concyclic.*

Why it works.

Given: quadrilateral $ABCD$ with $\angle BAD + \angle BCD = 180°$.
To show: $ABCD$ is cyclic.

1. Draw the circle through A, B and D (Theorem 5).
2. If C were outside it, P6 gives $\angle BED = \angle BCD$ and $\angle BED > \angle BCD$ together, which is impossible. The case of C inside is similar.
3. So C lies on the circle, and $ABCD$ is cyclic.

Things to Remember

Opposite angles of a cyclic quadrilateral add up to 180°. If opposite angles of a quadrilateral add up to 180°, it is cyclic.
If a segment subtends equal angles at two points on the same side of it, the four points are concyclic.

Example 2. $ABCD$ is a cyclic quadrilateral. AB is a diameter of the circle, and $\angle ADC = 120°$. Find $\angle BAC$.

FIG

Given: $ABCD$ cyclic, AB a diameter, $\angle ADC = 120°$.
To find: $\angle BAC$.

1. $\angle ABC = 180° - 120° = 60°$ (opposite angles of a cyclic quadrilateral, Theorem 11).
2. $\angle ACB = 90°$ (angle in a semicircle).
3. In $\triangle ABC$, $\angle BAC = 180° - 90° - 60° = 30°$.

Answer: $\angle BAC = 30°$.

Draw a rough figure for every question in Exercise 4A.

Exercise 4A

1. $ABCD$ is a cyclic quadrilateral. If $\angle A = 75°$, find $\angle C$. If $\angle B = 110°$, find $\angle D$.
2. $PQRS$ is a cyclic quadrilateral with $\angle P = (2x + 10)°$ and $\angle R = (3x - 20)°$. Find x, $\angle P$ and $\angle R$.
3. A quadrilateral $ABCD$ has $\angle A = 80°$, $\angle B = 110°$, $\angle C = 100°$ and $\angle D = 70°$. Can it be cyclic? Explain.
4. Shreya says, "Quadrilateral $ABCD$ has $\angle A = 100°$, $\angle B = 80°$, $\angle C = 100°$ and $\angle D = 80°$. Its angles $100°$ and $80°$ add up to $180°$, so it is cyclic." Find her mistake.
5. Four trees P, Q, R and S stand at the corners of a quadrilateral plot, with $\angle P = 75°$, $\angle Q = 95°$ and $\angle R = 105°$. A gardener wants to lay a circular path through all four trees. Is it possible? Explain.
6. Show that a rectangle is the only parallelogram that can be cyclic.
7. A rectangle is inscribed in a circle. Show that its diagonals meet at the centre of the circle.
8. $ABCD$ is a cyclic quadrilateral, and E is a point on CD extended beyond D. Explain why $\angle ADE = \angle ABC$.
9. $ABCD$ is a trapezium with $AB \parallel DC$ and $AD = BC$. Show that $ABCD$ is cyclic.
10. In P6, suppose C is *inside* the circle through A, B and D. Extend DC to meet the circle at E. Show that this is also impossible.
11. A cyclic quadrilateral has sides 5, 5, 12 and 12 units. Find its area.
12. Draw a circle of radius 4 cm and mark four points on it to form a quadrilateral. Measure its four angles and check Theorem 11. Then draw a quadrilateral whose vertices are *not* all on one circle, and check whether its opposite angles add up to $180°$.

4 Further questions

F1. $ABCD$ is a cyclic quadrilateral whose diagonals meet at P. $\angle DBC = 35°$ and $\angle BAC = 50°$. Find $\angle BAD$ and $\angle BCD$.

F2. ABC is a triangle, and BE and CF are its altitudes. Show that B, C, E and F are concyclic.

5 The twelve theorems of the chapter

Class	Our no.	NCERT no.	Statement
1	1	2	Equal chords subtend equal angles at the centre.
1	2	3	Chords that subtend equal angles at the centre are equal.
1	3	4	The line from the centre to the midpoint of a chord is perpendicular to the chord.
1	4	5	The perpendicular from the centre to a chord bisects the chord.
2	5	1	There is a unique circle through three non-collinear points.
2	6	6	Equal chords are equidistant from the centre.
2	7	7	Chords equidistant from the centre are equal.
2	8	8	Of two chords, the longer one is nearer the centre.
3	9	9	The angle at the centre is double the angle at a point on the rest of the circle. *Corollary:* the angle in a semicircle is $90°$.
4	10	10	If AB subtends equal angles at C and D on the same side, then A, B, C, D are concyclic.
4	11	11	Opposite angles of a cyclic quadrilateral add up to $180°$.
4	12	12	If opposite angles of a quadrilateral add up to $180°$, it is cyclic.